IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-095

1. Topic (12 words max)

Intelligent Project Management System for Undergraduate Students

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Undergraduate projects are an essential component of higher education, assisting students in developing their technical proficiency, teamwork abilities, and practical experience. Effective project management is still a problem in many colleges, though. According to Oakley et al. (2004), project teams are frequently constituted at random or on the basis of friendships rather than complimentary talents, which results in unbalanced teams and ineffective collaboration.

The free-rider problem, in which some students do less but obtain the same grade as their teammates, is one of the most prevalent problems in group assignments. According to Aggarwal and O'Brien (2008), this leads to unfair assessment procedures and lowers motivation among students who actively participate. Additionally, a lack of experience with time management, effort estimation, and technical feasibility assessment makes project planning difficult for many students. This frequently leads to impractical timetables, pressure at the last minute, and incomplete or subpar project outputs (Kerzner, 2017).

From the supervisors' point of view, keeping an eye on several project groups is mostly labor-intensive and manual. As the number of students increases, it becomes more challenging to monitor individual contributions, spot dangers early, and give prompt feedback (Fidalgo-Blanco et al., 2015). Project management solutions currently in use are mostly made for professional settings and do not adequately address educational needs including guided mentorship, fair contribution monitoring, and adaptive learning support.

In order to address these issues, this study suggests an Intelligent Project Management System that incorporates artificial intelligence for contribution analysis, skill-based team formation, project planning, and progress tracking. To directly support Sustainable Development Goal 4: Quality Education, the suggested approach seeks to enhance fairness, transparency, and learning outcomes in undergraduate project-based learning.

References

Aggarwal, P., & O'Brien, C. L. (2008). Social loafing on group projects. Journal of Marketing Education, 30(3), 255–264.

Fidalgo-Blanco, Á., Sein-Echaluce, M. L., & García-Peñalvo, F. J. (2015). Enhancing learning through project-based learning. Computers in Human Behavior, 51, 872–881.

Kerzner, H. (2017). Project Management: A Systems Approach to Planning, Scheduling, and Controlling. Wiley.

Oakley, B., Felder, R. M., Brent, R., & Elhajj, I. (2004). Turning student groups into effective teams. Journal of Student-Centered Learning, 2(1), 9–34.

6. Brief description of Existing Research and Systems (300 words max)

Summary of existing products/systems: Current solutions rely on disjointed tools: Learning Management Systems (LMS) for submissions, static tools like Trello/Jira for task management, and simple plagiarism checkers (Turnitin).

Summary of existing research: Existing research in Project Management focused on industry-level Agile methodologies which are too complex for students. Research on team formation often uses basic randomization or GPA matching without analyzing the specific semantic requirements of the chosen research topic.

Gaps/Limitations identified:

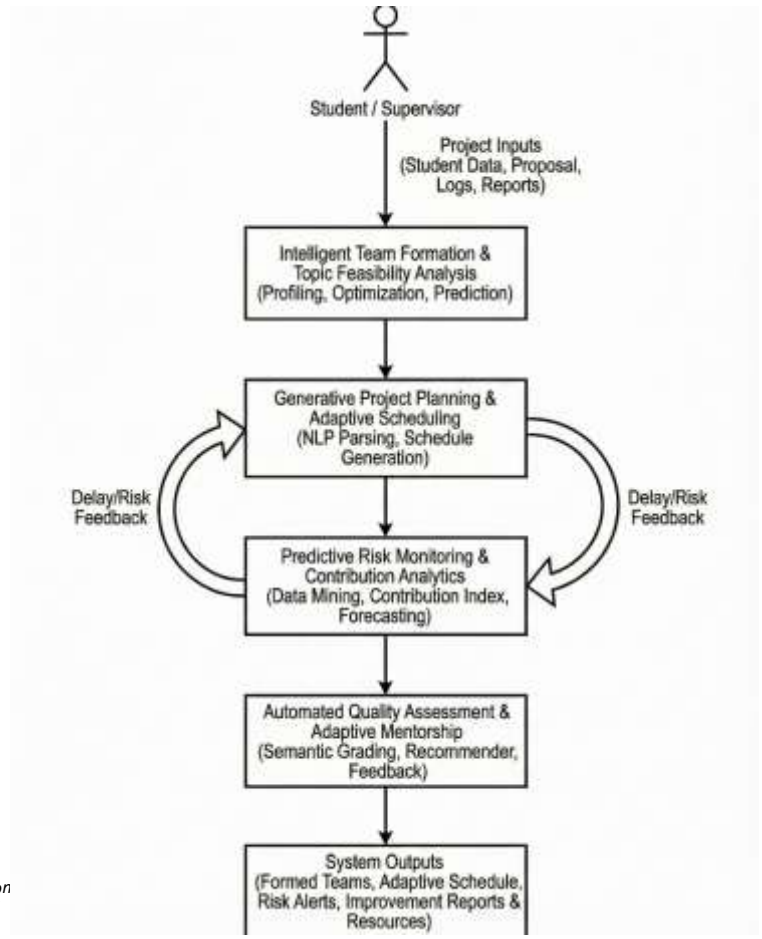
1. **Cold Start Problem:** Existing systems do not validate if a team actually possesses the specific skills required for their proposed topic before approval.
2. **Static Planning:** Tools like MS Project are passive; they do not auto-generate plans from a proposal text or self-correct when deadlines are missed.
3. **Lack of Quantitative Fairness:** There is no automated link between code contribution (GitHub/Jira) and academic grading to detect free-riders.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

How does this system/research address gaps identified in Section 6 above? The solution is an intelligent ecosystem that automates the lifecycle of a research project. It uses NLP to match students to topics based on skill vectors (solving the Cold Start problem), Generative AI to parse proposals into Gantt charts (solving Static Planning), and Machine Learning to forecast risk and calculate contribution indices (solving the Fairness gap).

System perspective of the Solution in the form of Input, Process, and Output:

- **Input:** Student academic profiles, Research Proposal text, GitHub development logs, Periodic reports.
- **Process:**
  - *Vector Profiling:* Quantifying skills and interests.
  - *Generative NLP:* Extracting WBS and Gantt charts from text.
  - *Predictive Analytics:* Forecasting project failure probability.
  - *Semantic Analysis:* Comparing report content against rubrics.
- **Output:** Optimized Team Configurations, Auto-generated/Adaptive Schedules, Contribution Indices (Fairness Score), and Automated Feedback Reports.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

How can members gain essential specialized domain expertise? Members will review literature on Natural Language Processing (BERT/GPT models), Multi-Objective Optimization algorithms (NSGA-II), and Predictive Risk Modeling.

How can a member gain the technical knowledge and skills required for the project? The team will utilize online courses (Coursera/Udemy) and documentation for Python (PyTorch/TensorFlow), GitHub APIs, and React/Node.js for system development.

How is the data required for the project collected?

- **Historical Data**: Anonymized past project proposals and grades (with permission).
- **Real-time Data**: Integration with live GitHub repositories and Jira boards during the testing phase.
- **Surveys**: Data regarding student preferences for team formation.

Does this project require ethical clearance? Yes, as the system analyzes student performance data and behavioral logs, ethical clearance will be sought to ensure privacy and data anonymity.

9. Objectives and Novelty

Main Objective To develop an intelligent, automated research project management ecosystem that optimizes team formation, generates adaptive project schedules, proactively monitors risk via predictive analytics, and provides automated, rubric-based academic assessment and mentorship.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Bandara NDM IT22114662	Intelligent Team Formation and Topic Feasibility Analysis	.***Student Profiling:*** *Develop an algorithm to quantify student skills, GPA, and research interests into vector profiles.*	The novelty lies in solving the "Cold Start" problem by coupling Team Formation with Feasibility Analysis. Unlike systems that just group students randomly, this model mathematically validates if the formed team actually has the skills to complete their specific chosen topic, ensuring viability before the project begins.
		2. **Optimization Engine:** Implement a multi-objective optimization algorithm to form groups that balance technical skills and personality traits.	

		3. **Feasibility Prediction:** Build a predictive model that scores the "Success Probability" of a chosen topic based on the team's combined skill set.	
Lekamge RLMD IT22117014	Generative Project Planning and Adaptive Scheduling	1. **Proposal Parsing:** Use NLP to extract key phases and deliverables from the unstructured text of the research proposal.	**Generative Planning with Self-Correction:** This moves beyond static calendar tools by using *Generative AI* to "read" a proposal and *create* the plan automatically. Its innovation is the *Adaptive Feedback Loop*: the schedule is not fixed; it intelligently recalculates milestones and meeting times in real-time based on actual team velocity, acting as an automated Project Manager.
		2. **Schedule Generation:** Train a model to auto-generate a technical Work Breakdown Structure (WBS) and Gantt chart tailored to the proposal's scope.	
		3. **Dynamic Adjustment:** Implement a "Self-Correcting" scheduler that automatically shifts deadlines when delays are detected.	
Kulathunga SAIK IT22109576	Predictive Risk Monitoring and Contribution Analytics	1. **Data Mining:** Integrate APIs (GitHub/Jira) to fetch development logs, commit history, and task updates	**Hybrid Risk & Fairness Modeling:**

			This component introduces *Quantitative Fairness* to group work. It uniquely combines *Project-Level Risk Prediction* (forecasting failure) with *Individual-Level Contribution Analytics*. This dual approach solves the "Free-Rider Problem" by using behavioral data to ensure individual accountability while simultaneously protecting the overall project health.
		2. **Contribution Engine:** Develop a fairness algorithm to calculate a "Contribution Index" for each member based on code volume and time spent.	
		3. **Risk Forecasting:** Train a Machine Learning model to predict "Project Failure" or "Delay" weeks in advance based on activity trends	
Rashmika RMD IT22101624	Automated Quality Assessment and Adaptive Mentorship	1. **Semantic Grading:** Implement an NLP pipeline to evaluate report structure, technical depth, and rubric compliance.	**Assessment-Driven Adaptive Learning:** The novelty is transforming "Grading" into "Mentorship." It uses NLP not just to score a paper, but to *diagnose* learning gaps. Unlike standard checkers, it closes

			the loop by immediately recommending specific learning resources (papers/code) to fix the detected errors, acting as a personal AI research mentor.
		2. **Resource Recommender:** Create a recommendation engine that suggests specific tutorials or papers based on identified report weaknesses	
		3. **Feedback Generation:** Automate the generation of detailed improvement reports for students.	
		4. **Grade Consistency Checking:** Cross-check grades with the student's previous academic records. High-performing students receiving unusually low marks are flagged for supervisor re-evaluation.	

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Prof	Samantha	Rajapaksha	
Co-Supervisor	Mrs.	Kaushika	Kahatapitiya	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				